

MA 403 Real Analysis

Homework 6

1. Let f be defined and continuous on a closed set $S \subset \mathbb{R}$. Let $A = \{x \in S : f(x) = 0\}$. Prove that A is a closed subset of $\mathbb{R}$.
2. Let f be continuous on a compact interval $[a, b]$. Suppose that f has a local maximum at x_1 and a local maximum at x_2 . Show that there must be a third point between x_1 and x_2 where f has a local minimum.
3. In each case, give an example of a function f , continuous on S and such that $f(S) = T$, or else explain why there can be no such f :
 - (a) $S = (0, 1)$, $T = (0, 1]$.
 - (b) $S = (0, 1)$, $T = (0, 1) \cup (1, 2)$.
 - (c) $S = \mathbb{R}$, $T = \mathbb{Q}$.
 - (d) $S = [0, 1] \times [0, 1]$, $T = \mathbb{R}^2$.
 - (e) $S = (0, 1) \times (0, 1)$, $T = \mathbb{R}^2$.
4. Let $f : (S, d_S) \rightarrow (T, d_T)$ be a function between two metric spaces. Prove that f is continuous on S if and only if $f(\overline{A}) \subseteq \overline{f(A)}$ for every subset A of S .
5. Prove that a metric space S is connected if and only if the only subsets of S which are both open and closed in S are the empty set and S itself.
6. Prove that if S is connected and if $S \subset T \subset \overline{S}$, then T is connected. In particular, this implies that the closure of a set is connected.
7. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = x^2$, prove that f is not uniformly continuous on $\mathbb{R}$.
8. Assume $f : (S, d_S) \rightarrow (T, d_T)$ is uniformly continuous on S . If $\{x_n\}$ is any Cauchy sequence in S , prove that $\{f(x_n)\}$ is a Cauchy sequence in T .

EXTRA:

9. Prove that the only connected subsets of $\mathbb{R}$ are (a) the empty set, (b) sets consisting of a single point, (c) intervals (open, closed, half-open, or infinite).